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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 1	
TITLE : Java Environment Setup and Basic Program Execution	

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyse outputs, and apply the concepts to solve practical programming problems.

Programme

Student.java

```java
public class Student {
    // Instance Variables
    int rollNo;
    String name;
    double marks;
    // Default Constructor
    Student() {
        rollNo = 0;
        name = "Not Assigned";
        marks = 0.0;
    }
    // Parameterized Constructor
    Student(int r, String n, double m) {
        rollNo = r;
        name = n;
        marks = m;
    }
    // Method to display student details
    void displayDetails() {
        System.out.println("Roll No : " + rollNo);
        System.out.println("Name    : " + name);
        System.out.println("Marks   : " + marks);
    }
    // Method to calculate grade
    String getGrade() {
        if (marks >= 75)
            return "A";
        else if (marks >= 60)
            return "B";
        else if (marks >= 40)
            return "C";
        else
            return "Fail";
    }
    public static void main(String[] args) {
        System.out.println("Student Using Default Constructor");
        Student s1 = new Student();
        s1.displayDetails();
        System.out.println("Grade : " + s1.getGrade());
        System.out.println();

        System.out.println("Student Using Parameterized Constructor");
        Student s2 = new Student(101, "Ombabu", 86.5);
        s2.displayDetails();
        System.out.println("Grade : " + s2.getGrade());
    }
}
```

Output

Output

```
Student Using Default Constructor
Roll No : 0
Name    : Not Assigned
Marks   : 0.0
Grade : Fail

Student Using Parameterized Constructor
Roll No : 101
Name    : Ombabu
Marks   : 86.5
Grade : A
```

Conclusion

This assignment helped in understanding the basic concepts of Java Programming, including the Java environment (JDK, JRE & JVM), variables, methods & constructors. It demonstrated how Java programs are compiled & executed, and how variables store data while methods organize program logic into reusable blocks. These concepts form the foundation of OOP, and are essential for developing efficient, reliable, and maintainable Java application

Questions

1. Explain the main concept used in Java environment setup, variables and methods.

→ Java Environment Setup : It involves installing the JDK, configuring environment variables, and verifying the installation. The JDK includes the compiler (javac), while the JRE provides runtime environment.

→ Variables : They are named memory location used to store data. Java supports different types of variables such as local, instance, static & parameter variables.

→ Methods : Methods are blocks of code that perform specific task. They improve code reusability, readability & modularity

2. What are the advantages of using Java variables and methods?

→ improve code readability & maintainablity
→ Enable data storage & manipulation
→ promote code reusability through method calls
→ Reduce code duplication.
→ Support modular programming
→ making debugging & testing easier

3. How can this implementation be improved?

→ Apply OOP Principles such as encapsulation & inheritance
→ Use meaningful variable & method names.
→ Add user input validation
→ Include comments & proper documentation
→ Implement exception handling for better error management.
→ Organize the Program into multiple classes & Packages for better maintainability

4. What are the real-world applications?

→ Banking & financial management System
→ Student information & library management System
→ Hospital & healthcare management software
→ Inventory & billing Systems
→ Enterprise business applications
→ Android mobile application development
→ Cloud-based & distributed software systems